

## Chapter 20 Homework Answer Sheet

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### Part A Multiple choice (15 points)

1-5 EABCB

6-10 DCDCB

11-15 AEEDA

### Part B Short questions (15 points)

#### 16. (7 points)

(a) (1 point) The half-cell with  $[\text{Zn}^{2+}] = 3.75 \times 10^{-4} \text{ M}$  is the anode.

(b) (2 points)  $E_{\text{cell}}^{\circ} = E_{\text{red}}^{\circ}(\text{cathode}) - E_{\text{red}}^{\circ}(\text{anode}) = (-0.76\text{V}) - (-0.76\text{V}) = 0$

(c) (3 points)

The overall cell reaction is  $\text{Zn}^{2+}(\text{aq, concentrated}) \rightarrow \text{Zn}^{2+}(\text{aq, dilute})$

$$E = E^{\circ} - \frac{0.0592}{n} \log Q = 0 - \frac{0.0592}{2} \log \frac{[\text{Zn}^{2+}]_{\text{dilute}}}{[\text{Zn}^{2+}]_{\text{concentrated}}} = -\frac{0.0592}{2} \log \frac{3.75 \times 10^{-4} \text{ M}}{1.35 \text{ M}} = 0.105\text{V}$$

(d) (1 point) Increase.

#### 17. (8 points)

(a) (2 points)  $\text{S}_2\text{O}_6^{2-}(\text{aq}) + 4 \text{H}^+(\text{aq}) + 2 \text{Fe}^{2+}(\text{aq}) \rightarrow 2 \text{H}_2\text{SO}_3(\text{aq}) + 2 \text{Fe}^{3+}(\text{aq})$

(b) (3 points)

$$E^{\circ} = E_{\text{red}}^{\circ}(\text{cathode}) - E_{\text{red}}^{\circ}(\text{anode}) = 0.60 - 0.77 = -0.17 \text{ V}$$

$$\Delta G^{\circ} = -nFE^{\circ} = -2 \times 96485 \text{ C/mol} \times (-0.17) \text{ J/C} = 3.3 \times 10^4 \text{ J/mol OR } 33 \text{ kJ/mol}$$

(c) (3 points)

$$\Delta G^{\circ} = -RT \ln K$$

$$3.3 \times 10^4 \text{ J/mol} = -(8.314 \text{ J/K mol})(298 \text{ K}) \ln K$$

$$\ln K = -13$$

$$K = 2 \times 10^{-6}$$